

# Fall 2025 CPTS530 Homework 03 Report

## Numerical Analysis

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## Question 1

**Problem Statement:** Suppose that  $x > 0$ ,  $y < 0$ . Using definition of absolute value and splitting into cases, show that  $|x + y| \leq |x - y|$ .

**Solution:**

**Given:**  $x > 0$ ,  $y < 0$

**To show:**  $|x + y| \leq |x - y|$

**Visual Representation:**

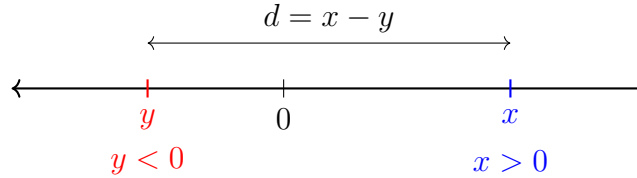


Figure 1: Number line showing the relationship between  $x$ ,  $y$  and  $x - y$

Since  $x > 0$  and  $y < 0$ , we can assume that  $x - y = d$  where  $d > 0$ .

Therefore,  $\text{RHS} = |x - y| = d$  is always true.

Now we need to consider cases for  $|x + y|$  based on whether  $x + y$  is positive or negative.

**Case I:**  $x + y \geq 0$

From  $x - y = d$ , we get  $x = y + d$ .

Therefore:  $x + y = (y + d) + y = 2y + d$

Since  $x + y \geq 0$ , we have  $|x + y| = x + y = 2y + d$ .

So  $\text{LHS} = 2y + d$  and  $\text{RHS} = d$ .

Since  $y < 0$ , we have  $2y < 0$ , which means  $\text{LHS} = 2y + d < d = \text{RHS}$ .

**Case II:**  $x + y < 0$

From  $x - y = d$ , we get  $y = x - d$ .

Therefore:  $x + y = x + (x - d) = 2x - d < 0$

Since  $x + y < 0$ , we have  $|x + y| = -(x + y) = -(2x - d) = d - 2x$ .

So  $\text{LHS} = d - 2x$  and  $\text{RHS} = d$ .

Since  $x > 0$ , we have  $2x > 0$ , which means  $\text{LHS} = d - 2x < d = \text{RHS}$ .

**Conclusion:**

Cases I and II exhaustively cover all possible scenarios. In both cases, we have shown that  $\text{LHS} < \text{RHS}$ , which means  $|x + y| < |x - y|$ .

Therefore, we have proven the stronger result:  $|x + y| \leq |x - y|$  for  $x > 0, y < 0$

**Hence Proved.**  $\square$

## Question 2

**Problem Statement:** Suppose you have an iterative algorithm for finding a fixed point of a function  $F(x)$  on a certain interval. Write a short text (2-4 paragraphs) explaining the following:

1. How do you decide when to stop iterating?
2. What is your strategy for making sure that the computed solution is close to the exact solution?

DO NOT use the standard methodology and estimates from the textbook. Try to find other ways of accomplishing the goals above. Discuss briefly the advantages and disadvantages of your error control approach.

**Solution:**

## Question 3

**Problem Statement:** Do Problem 23 c) from Section 3.4.

**Problem 3.4.23 c):** Find the order of convergence of the sequence:

$$x_n = \left(1 + \frac{1}{n}\right)^{\frac{1}{2}}$$

**Solution:**

## Question 4

**Problem Statement:** Do Problem 32 from Section 3.4.

**Problem 3.4.32:** Let  $\frac{1}{2} \leq q \leq 1$ , and define  $F(x) = 2x - qx^2$ . On what interval can it be guaranteed that the method of iteration using  $F$  will converge to a fixed point? (This problem is related to Problem 3.2.5, p. 90.)

**Related Problem 3.2.5:** What is the purpose of the following iteration formula?

$$x_{n+1} = 2x_n - x_n^2 y$$

Identify it as the Newton iteration for a certain function.

**Solution:**

### Analysis of Related Problem 3.2.5

First, let's identify what the iteration formula  $x_{n+1} = 2x_n - x_n^2 y$  represents as a Newton iteration.

The Newton-Raphson method for solving  $f(x) = 0$  has the form:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Comparing with our given iteration:

$$x_{n+1} = 2x_n - yx_n^2 = x_n - (-x_n + yx_n^2)$$

Therefore:

$$\frac{f(x_n)}{f'(x_n)} = yx_n^2 - x_n$$

To solve for  $f(x_n)$  we'll have to use integration. Let's temporarily switch to  $x$  as a variable instead of  $x_n$ .

$$\frac{f(x)}{f'(x)} = yx^2 - x$$

or,

$$\frac{f(x)}{\frac{df(x)}{dx}} = yx^2 - x$$

Cross-multiplying with  $\frac{1}{dx}$  on both sides:

$$\frac{f(x)}{df(x)} = \frac{yx^2 - x}{dx}$$

Equivalently, we can write:

$$\frac{df(x)}{f(x)} = \frac{dx}{yx^2 - x}$$

Integrating both sides:

$$\ln |f(x)| = \int \frac{dx}{yx^2 - x} = \int \frac{dx}{x(yx - 1)}$$

Dividing numerator and denominator or RHS by  $y$ :

$$\ln |f(x)| = \int \frac{1/y}{x(x - 1/y)} dx$$

Therefore:

$$\ln |f(x)| = \int \left( \frac{1}{x - 1/y} - \frac{1}{x} \right) dx$$

$$\text{or, } \ln |f(x)| = \ln \left| x - \frac{1}{y} \right| - \ln |x| + \ln k, \quad k \neq 0$$

$$\text{or, } \ln |f(x)| = \ln \left| \frac{k(x - \frac{1}{y})}{x} \right|$$

Taking the exponential:

$$f(x) = \frac{k}{x} \left( x - \frac{1}{y} \right)$$

Giving us the root of  $f(x)$  or alternatively the fixed point of  $F(x)$  at

$$x = r = \frac{1}{y}$$

## Main Problem Solution

Now for the main problem with  $F(x) = 2x - qx^2$  where  $\frac{1}{2} \leq q \leq 1$ .

Basically here  $y = q$ , so the fixed point is  $r = \frac{1}{q}$ .

For the fixed point iteration to converge, we need  $|F'(x)| < 1$  in an interval  $[a, b]$  containing the fixed point  $r = \frac{1}{q}$ .

**Step 1: Find the derivative**

$$F'(x) = 2 - 2qx$$

**Step 2: Applying the convergence condition (from a previous theorem in homework 2)** For convergence:  $|F'(x)| < 1$

This gives us:  $|2 - 2qx| < 1$

**Step 3: Solve the inequality**

$$-1 < 2 - 2qx < 1$$

For upper limit:  $2 - 2qx > -1$

$$2qx < 3$$

$$x < \frac{3}{2q}$$

From lower limit:  $2 - 2qx < 1$

$$2qx > 1$$

$$x > \frac{1}{2q}$$

Thus:  $\frac{1}{2q} < x < \frac{3}{2q}$ . We may note that the fixed point  $r = \frac{1}{q}$  lies in this interval as well.

Normally, a contractive function requires a closed interval like  $[a, b]$ , but here we have an open interval  $(a, b)$ .

This does not mean that the extreme points  $a$  and  $b$  are necessarily ruled out. We can check for  $F(a)$  and  $F(b)$  to see if they 'get' mapped into the contractive region and if they both are, we can safely claim that the closed interval  $[a, b]$  is also valid.

Calculating  $F(a)$ :

$$F(a) = F\left(\frac{1}{2q}\right) = 2\left(\frac{1}{2q}\right) - q\left(\frac{1}{2q}\right)^2 = \frac{1}{q} - \frac{1}{4q} = \frac{3}{4q}$$

Calculating  $F(b)$ :

$$F(b) = F\left(\frac{3}{2q}\right) = 2\left(\frac{3}{2q}\right) - q\left(\frac{3}{2q}\right)^2 = \frac{3}{q} - \frac{9}{4q} = \frac{3}{4q}$$

Thus we can see that both  $F(a)$  and  $F(b)$  map to  $\frac{3}{4q}$  which lies in the interval  $(a, b)$ .

Therefore we can safely say that the closed interval  $[a, b] = \left[\frac{1}{2q}, \frac{3}{2q}\right]$  is also valid.

Now as  $q$  itself is given as a parameter in the range  $\frac{1}{2} \leq q \leq 1$ , we can see that the interval  $\left[\frac{1}{2q}, \frac{3}{2q}\right]$  will vary based on the value of  $q$ .

We want to calculate the largest 'common' interval of  $[a, b]$  valid for all  $q$  in the given range.

Actually this answer depends on whether we have an apriori knowledge of  $q$  or not.

If we do know the value of  $q$  beforehand, then we can use the interval  $\left[\frac{1}{2q}, \frac{3}{2q}\right]$  for that specific  $q$ .

However if  $q$  is strictly to be treated as a parameter between  $\frac{1}{2}$  and 1, we're interested in some 'universal' interval where  $F(x)$  will converge to a fixed point regardless of the value of  $q$  – also it may be noted that the fixed point itself is allowed to lie outside  $[a, b]$ .

We recognize that the values for  $a$  and  $b$  are monotonically dependent on the value of  $q$ , both  $a$  and  $b$  decrease as  $q$  increases.

We can then exploit this monotonicity to come up with a formula for the universal interval of convergence.

### Step 2: Intersection of all these intervals

The intersection of all  $I_q$  over  $q \in [0.5, 1]$  is:

$$\bigcap_{q \in [0.5, 1]} I_q = [\max a(q), \min b(q)] = [a(0.5), b(1)] = \left[1, \frac{3}{2}\right]$$

Thus the largest common interval for all  $q$  in the range  $\frac{1}{2} \leq q \leq 1$  is  $\left[1, \frac{3}{2}\right]$ .

**Final Answer:**

The method of iteration using  $F(x) = 2x - qx^2$  will converge to the fixed point  $r = \frac{1}{q}$  on the interval:

**If  $q$  is known apriori:**

$$\left[ \frac{1}{2q}, \frac{3}{2q} \right]$$

**If  $q$  is strictly a parameter in the range  $\frac{1}{2} \leq q \leq 1$ :**

$$\left[ 1, \frac{3}{2} \right]$$

□